Importing Necessary Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
import xgboost as xgb
from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split, GridSearchCV, cross_val_
score
```

Reading Training Dataset

In [2]:
```
data_train = pd.read_csv('data_train.csv')
data_train.head()
```

Out[2]:

	ID	Unnamed: 0	RA_ICRS	DE_ICRS	Source	Plx	PM	pmRA	pmDE	Gmag
0	1	1	44.375187	2.895901	1.400000e+15	1.8937	16.954	7.031	-15.427	10.369156
1	2	2	42.959304	2.920459	1.550000e+15	2.9732	2.528	1.795	1.780	10.038978
2	3	3	44.031079	3.203683	1.610000e+15	2.2337	10.776	-3.151	-10.305	9.810181
3	4	4	45.463599	3.138095	1.750000e+15	4.0179	10.053	1.828	9.885	8.163041
4	5	5	43.784164	3.486814	2.060000e+15	11.6458	32.355	29.688	-12.863	6.855547

5 rows × 29 columns

In [3]:
```
data_train.shape
```

Out[3]: (148538, 29)

In [4]:
```
data_train.dtypes
```

Out[4]:
```
ID              int64
Unnamed: 0      int64
RA_ICRS       float64
DE_ICRS       float64
Source        float64
Plx           float64
PM            float64
pmRA          float64
pmDE          float64
Gmag          float64
e_Gmag        float64
BPmag         float64
e_BPmag       float64
RPmag         float64
e_RPmag       float64
GRVSmag       float64
e_GRVSmag     float64
BP-RP         float64
BP-G          float64
G-RP          float64
pscol         float64
Teff          float64
Dist          float64
Rad           float64
Lum-Flame     float64
Mass-Flame    float64
Age-Flame     float64
z-Flame       float64
SpType-ELS     object
dtype: object
```

Finding Missing Values

In [5]:
```
data_train.isnull().sum()
```

Out[5]:
```
ID                 0
Unnamed: 0         0
RA_ICRS            0
DE_ICRS            0
Source             0
Plx                0
PM                 0
pmRA               0
pmDE               0
Gmag               0
e_Gmag             0
BPmag              0
e_BPmag            0
RPmag              0
e_RPmag            0
GRVSmag        64054
e_GRVSmag      64054
BP-RP              0
BP-G               0
G-RP               0
pscol         143685
Teff               0
Dist               0
Rad                0
Lum-Flame       2995
Mass-Flame     12551
Age-Flame      38016
z-Flame         2995
SpType-ELS         0
dtype: int64
```

Handling Missing Values

Dropping Columns with large number of missing values

In [6]:
```
data_train = data_train.drop(columns=['GRVSmag','e_GRVSmag','pscol'])
```

Imputing missing values with median strategy

In [7]:
```
from sklearn.impute import SimpleImputer

impute_columns = ['Lum-Flame','Mass-Flame','Age-Flame','z-Flame']
imputer = SimpleImputer(strategy='median')
data_train[impute_columns] = imputer.fit_transform(data_train[impute_columns])
```

In [8]:
```
data_train.isnull().sum()
```

Out[8]:
```
ID            0
Unnamed: 0    0
RA_ICRS       0
DE_ICRS       0
Source        0
Plx           0
PM            0
pmRA          0
pmDE          0
Gmag          0
e_Gmag        0
BPmag         0
e_BPmag       0
RPmag         0
e_RPmag       0
BP-RP         0
BP-G          0
G-RP          0
Teff          0
Dist          0
Rad           0
Lum-Flame     0
Mass-Flame    0
Age-Flame     0
z-Flame       0
SpType-ELS    0
dtype: int64
```

Encoding Categorical Variable SpType-ELS

In [9]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
data_train['SpType-ELS'] = le.fit_transform(data_train['SpType-ELS'])
```

In [10]:
```
data_train['SpType-ELS'].unique()
```

Out[10]:
```
array([0, 1])
```

Splitting the data into features(X) and target variable(y)

In [11]:
```
X = data_train.drop(columns=['ID','Unnamed: 0','SpType-ELS'])
y = data_train['SpType-ELS']
```

Splitting the data into training and validation set

In [12]:
```
from sklearn.model_selection import train_test_split
X_train, X_val, y_train, y_val = train_test_split(X,y, test_size=0.2, random_s
tate=42)
```

In [13]:
```
print('X_train shape: ',X_train.shape)
print('X_val shape: ',X_val.shape)
```

```
X_train shape:  (118830, 23)
X_val shape:  (29708, 23)
```

Standardizing the Features

In [14]:
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_val = scaler.transform(X_val)
```

Modelling

Logistic Regression

In [15]:
```
lr = LogisticRegression(random_state=42,max_iter=1000)
```

In [16]:
```
#Training the classifier
lr.fit(X_train,y_train)
```

Out[16]:
```
▾ LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

In [17]:
```
#Predicting on the validation set
y_pred_lr = lr.predict(X_val)
```

In [18]:
```
#Evaluating the Classifier
accuracy_lr = accuracy_score(y_val,y_pred_lr)
print("Logistic Regression Validation Accuracy:", accuracy_lr*100, '%')
print()
print(classification_report(y_val,y_pred_lr))
```

```
Logistic Regression Validation Accuracy: 99.3200484717921 %

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.99      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Applying Hyperparameter Tunning on Logistic Regression Model

In [19]:
```
#Defining the hyperparameters grid for Logistic Regression
lr_param_grid = {
    'C': [0.01, 0.1, 1, 10, 100],
    'solver': ['liblinear']
}
```

In [20]:
```
lr_grid_search = GridSearchCV(estimator=LogisticRegression(random_state=42),
                              param_grid=lr_param_grid, cv=5, scoring='accurac
y', n_jobs=-1)

#Training the model with GridSearchCV
lr_grid_search.fit(X_train, y_train)
best_lr_classifier = lr_grid_search.best_estimator_

#Predicting and evaluating
y_pred_lr = best_lr_classifier.predict(X_val)
accuracy_lr = accuracy_score(y_val, y_pred_lr)
print("Logistic Regression Validation Accuracy After Tuning:", accuracy_lr*10
0, '%')
print("Best Hyperparameters:", lr_grid_search.best_params_)
print(classification_report(y_val, y_pred_lr))
```

```
Logistic Regression Validation Accuracy After Tuning: 99.35034334186078 %
Best Hyperparameters: {'C': 100, 'solver': 'liblinear'}
              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.99      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Cross-Validation to check performance

In [21]:
```
lr_cv_scores = cross_val_score(best_lr_classifier, X_train,y_train,cv=5,scorin
g='accuracy')
print("Logistic Regression Cross-Validation Mean Accuracy", np.mean(lr_cv_scor
es)*100,'%')
```

```
Logistic Regression Cross-Validation Mean Accuracy 99.34780779264494 %
```

KNeighbors Classifier (KNN)

In [22]:
```
knn = KNeighborsClassifier()
knn.fit(X_train,y_train)
```

Out[22]:
```
▾ KNeighborsClassifier
KNeighborsClassifier()
```

In [23]:
```
y_pred_knn = knn.predict(X_val)
```

In [24]:
```
accuracy_knn = accuracy_score(y_val,y_pred_knn)
print("KNeighbor Classifier Validation Accuracy:", accuracy_knn*100, '%')
print()
print(classification_report(y_val,y_pred_knn))
```

```
KNeighbor Classifier Validation Accuracy: 98.83196445401911 %

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.98      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Applying Hyperparameter Tunning on KNeighbor Classifier

In [25]:
```
knn_param_grid = {
  'n_neighbors' : [3,5,7,9,11],
  'weights' : ['uniform','distance']
}
```

In [26]:
```
knn_grid_search = GridSearchCV(estimator=KNeighborsClassifier(), param_grid=kn
n_param_grid, cv=5, scoring='accuracy',n_jobs=-1)
```

In [27]:
```
knn_grid_search.fit(X_train,y_train)
best_knn_classifier = knn_grid_search.best_estimator_
```

In [28]:
```
y_pred_knn = best_knn_classifier.predict(X_val)
accuracy_knn_new = accuracy_score(y_val,y_pred_knn)
print('KNeighbors Classifier Validation Accuracy', accuracy_knn_new*100,'%')
print()
print('Best Hyperparameter:',knn_grid_search.best_params_)
print()
print(classification_report(y_val,y_pred_knn))
```

```
KNeighbors Classifier Validation Accuracy 98.83869664736771 %

Best Hyperparameter: {'n_neighbors': 5, 'weights': 'distance'}

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.98      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Cross-Validation

In [29]:
```
knn_cv_scores = cross_val_score(best_knn_classifier,X_train,y_train, cv=5, sco
ring='accuracy')
print("KNeighbors Classifier Cross-Validation Mean Accuracy:", np.mean(knn_cv_
scores)*100,'%')
```

```
KNeighbors Classifier Cross-Validation Mean Accuracy: 98.86981402002861 %
```

Random Forest Classifier

In [30]:
```
rf = RandomForestClassifier(random_state=42)
rf.fit(X_train,y_train)
```

Out[30]:
```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

In [31]:
```
y_pred_rf = rf.predict(X_val)
```

In [32]:
```
accuracy_rf = accuracy_score(y_val,y_pred_rf)
print("Random Forest Classifier Validation Accuracy:", accuracy_rf*100, '%')
print()
print(classification_report(y_val,y_pred_rf))
```

```
Random Forest Classifier Validation Accuracy: 99.65665813922175 %

              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16044
           1       0.99      1.00      1.00     13664

    accuracy                           1.00     29708
   macro avg       1.00      1.00      1.00     29708
weighted avg       1.00      1.00      1.00     29708
```

Applying Hyperparameter Tuning on Random Forest Classifier

In [33]:
```
rf_param_grid = {
    'max_depth': [None,10,20],
    'min_samples_split': [2,5,10]
}
```

In [34]:
```
rf_grid_search = GridSearchCV(estimator=RandomForestClassifier(random_state=4
2),param_grid=rf_param_grid,cv=5,scoring='accuracy',n_jobs=-1)
```

In [35]:
```
rf_grid_search.fit(X_train,y_train)
best_rf_classifier = rf_grid_search.best_estimator_
```

In [38]:
```
y_pred_rf = best_rf_classifier.predict(X_val)
accuracy_rf_new = accuracy_score(y_val,y_pred_rf)
print('Random Forest Classifier Validation Score:', accuracy_rf_new*100,'%')
print()
print("Best Hyperparameter:", rf_grid_search.best_params_)
print()
print(classification_report(y_val,y_pred_rf))
```

```
Random Forest Classifier Validation Score: 99.65329204254746 %

Best Hyperparameter: {'max_depth': 20, 'min_samples_split': 2}

              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16044
           1       0.99      1.00      1.00     13664

    accuracy                           1.00     29708
   macro avg       1.00      1.00      1.00     29708
weighted avg       1.00      1.00      1.00     29708
```

Cross-Validation

In [39]:
```
rf_cv_scores = cross_val_score(best_rf_classifier,X_train,y_train, cv=5, scori
ng='accuracy')
print("Random Forest Cross Validation Mean Score:", np.mean(rf_cv_scores)*10
0,'%')
```

```
Random Forest Cross Validation Mean Score: 99.65833543717918 %
```

Decision Tree Classifier

In [40]:
```
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train,y_train)
```

Out[40]:
```
▾ DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

In [41]:
```
y_pred_dt = dt.predict(X_val)
```

In [42]:
```
accuracy_dt = accuracy_score(y_val,y_pred_dt)
print('Decision Tree Classifier Validation Score:', accuracy_dt*100,'%')
print()
print(classification_report(y_val,y_pred_dt))
```

```
Decision Tree Classifier Validation Score: 99.39746869530093 %

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.99      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Applying Hyperparameter Tuning on Decision Tree Classifier

In [43]:
```
dt_param_grid = {
    'max_depth' : [3,5,7,10],
    'min_samples_split' : [2,5,10]
}
```

In [44]:
```
dt_grid_search = GridSearchCV(estimator=DecisionTreeClassifier(random_state=4
2),param_grid=dt_param_grid,cv=5,scoring='accuracy',n_jobs=-1)
```

In [45]:
```
dt_grid_search.fit(X_train,y_train)
best_dt_classifier = dt_grid_search.best_estimator_
```

In [46]:
```
y_pred_dt = best_dt_classifier.predict(X_val)
accuracy_dt_new = accuracy_score(y_val,y_pred_dt)
print('Decision Tree Classifier Validation Score:', accuracy_dt_new*100,'%')
print()
print("Best Hyperparameter:", dt_grid_search.best_params_)
print()
print(classification_report(y_val,y_pred_dt))
```

```
Decision Tree Classifier Validation Score: 99.57923791571294 %

Best Hyperparameter: {'max_depth': 7, 'min_samples_split': 2}

              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16044
           1       0.99      1.00      1.00     13664

    accuracy                           1.00     29708
   macro avg       1.00      1.00      1.00     29708
weighted avg       1.00      1.00      1.00     29708
```

Cross-Validation

In [47]:
```
dt_cv_scores = cross_val_score(best_dt_classifier, X_train, y_train, cv=5, sco
ring='accuracy')
print("Decision Tree Classifier Mean Validation Accuracy:", np.mean(dt_cv_scor
es)*100,'%')
```

```
Decision Tree Classifier Mean Validation Accuracy: 99.5817554489607 %
```

Support Vector Classifier

In [48]:
```
svc = SVC(random_state=42)
svc.fit(X_train,y_train)
```

Out[48]:
```
▾ SVC
SVC(random_state=42)
```

In [49]:
```
y_pred_svc = svc.predict(X_val)
```

In [50]:
```
accuracy_svc = accuracy_score(y_val,y_pred_svc)
print("Support Vector Classifier Validation Accuracy:", accuracy_svc*100, '%')
print()
print(classification_report(y_val,y_pred_svc))
```

```
Support Vector Classifier Validation Accuracy: 99.4243974686953 %

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.99      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Applying Hyperparameter Tunning for Support Vector Classifier

In [51]:
```
svc_param_grid = {
    'gamma' : [1,0.1,0.01],
    'kernel' : ['linear','rbf']
}
```

In [52]:
```
svc_grid_search = GridSearchCV(estimator=SVC(random_state=42),param_grid=svc_p
aram_grid,cv=5,scoring='accuracy',n_jobs=-1)
```

In [53]:
```
svc_grid_search.fit(X_train,y_train)
best_svc_classifier = svc_grid_search.best_estimator_
```

In [54]:
```
y_pred_svc = best_svc_classifier.predict(X_val)
accuracy_svc_new = accuracy_score(y_val,y_pred_svc)
print('Support Vector Classifier Validation Score:', accuracy_svc_new*100,'%')
print()
print("Best Hyperparameter:", svc_grid_search.best_params_)
print()
print(classification_report(y_val,y_pred_svc))
```

```
Support Vector Classifier Validation Score: 99.4176652753467 %

Best Hyperparameter: {'gamma': 0.1, 'kernel': 'rbf'}

              precision    recall  f1-score   support

           0       0.99      0.99      0.99     16044
           1       0.99      0.99      0.99     13664

    accuracy                           0.99     29708
   macro avg       0.99      0.99      0.99     29708
weighted avg       0.99      0.99      0.99     29708
```

Cross-Validation

In [55]:
```
svc_cv_scores = cross_val_score(best_svc_classifier, X_train, y_train, cv=5, s
coring='accuracy')
print("Support Vector Classifier Mean Validation Accuracy:", np.mean(svc_cv_sc
ores)*100,'%')
```

```
Support Vector Classifier Mean Validation Accuracy: 99.42438778086343 %
```

XGBoost Classifier

In [56]:
```
xg = xgb.XGBClassifier(random_state=42)
xg.fit(X_train,y_train)
```

Out[56]:
```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=No
ne,
              enable_categorical=False, eval_metric=None, feature_types=No
ne,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=No
ne,
              max_cat_threshold=None, max_cat_to_onehot=None,
```

In [57]:
```
y_pred_xg = xg.predict(X_val)
```

In [58]:
```
accuracy_xg = accuracy_score(y_val,y_pred_xg)
print("XGBoost Classifier Validation Score:", accuracy_xg*100,'%')
print()
print(classification_report(y_val,y_pred_xg))
```

```
XGBoost Classifier Validation Score: 99.69031910596472 %

              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16044
           1       1.00      1.00      1.00     13664

    accuracy                           1.00     29708
   macro avg       1.00      1.00      1.00     29708
weighted avg       1.00      1.00      1.00     29708
```

Applying Hyperparameter Tunning for XGBoost Classifier

In [59]:
```
xgb_param_grid = {
    'n_estimators': [50,100,200],
    'learning_rate': [0.01,0.1,0.2],
}
```

In [60]:
```
xgb_grid_search = GridSearchCV(estimator=xgb.XGBClassifier(random_state=42),pa
ram_grid=xgb_param_grid,cv=5,scoring='accuracy',n_jobs=-1)
```

In [61]:
```
xgb_grid_search.fit(X_train,y_train)
best_xgb_classifier = xgb_grid_search.best_estimator_
```

In [62]:
```
y_pred_xgb = best_xgb_classifier.predict(X_val)
accuracy_xgb_new = accuracy_score(y_val,y_pred_xgb)
print('XGBoost Classifier Validation Score:', accuracy_xgb_new*100,'%')
print()
print("Best Hyperparameter:", xgb_grid_search.best_params_)
print()
print(classification_report(y_val,y_pred_xgb))
```

```
XGBoost Classifier Validation Score: 99.68695300929042 %

Best Hyperparameter: {'learning_rate': 0.2, 'n_estimators': 200}

              precision    recall  f1-score   support

           0       1.00      1.00      1.00     16044
           1       1.00      1.00      1.00     13664

    accuracy                           1.00     29708
   macro avg       1.00      1.00      1.00     29708
weighted avg       1.00      1.00      1.00     29708
```

Cross-Validation

In [63]:
```
xgb_cv_scores = cross_val_score(best_xgb_classifier, X_train, y_train, cv=5, s
coring='accuracy')
print("XGBoost Classifier Mean Validation Score:", np.mean(xgb_cv_scores)*10
0,'%')
```

```
XGBoost Classifier Mean Validation Score: 99.69031389379786 %
```

Comparing Results and Selecting the Best Model

In [69]:
```
print(f"Logistic Regression Classifier Validation Accuracy: {accuracy_lr*100:.
2f} %")
print(f"KNeighbors Classifier Validation Accuracy: {accuracy_knn_new*100:.2f}
%")
print(f"Random Forest Classifier Validation Accuracy: {accuracy_rf_new*100:.2
f} %")
print(f"Decision Tree Classifier Validation Accuracy: {accuracy_dt_new*100:.2
f} %")
print(f"Support Vector Classifier Validation Accuracy: {accuracy_svc_new*100:.
2f} %")
print(f"XGBoost Classifier Validation Accuracy: {accuracy_xgb_new*100:.2f} %")
```

```
Logistic Regression Classifier Validation Accuracy: 99.35 %
KNeighbors Classifier Validation Accuracy: 98.84 %
Random Forest Classifier Validation Accuracy: 99.65 %
Decision Tree Classifier Validation Accuracy: 99.58 %
Support Vector Classifier Validation Accuracy: 99.42 %
XGBoost Classifier Validation Accuracy: 99.69 %
```

The Best Model

Finding best model

In [70]:
```
best_model = max([(accuracy_lr,best_lr_classifier),
                  (accuracy_knn_new,best_knn_classifier),
                  (accuracy_rf_new,best_rf_classifier),
                  (accuracy_dt_new,best_dt_classifier),
                  (accuracy_svc_new,best_svc_classifier),
                  (accuracy_xgb_new,best_xgb_classifier)], key=lambda x:x[0])
[1]
best_model
```

Out[70]:

XGBClassifier

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=No
ne,
              enable_categorical=False, eval_metric=None, feature_types=No
ne,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.2, max_bin=Non
e,
              max_cat_threshold=None, max_cat_to_onehot=None,
```

ROC Curve & AUC Analysis

Predicting Probabilities of the best model

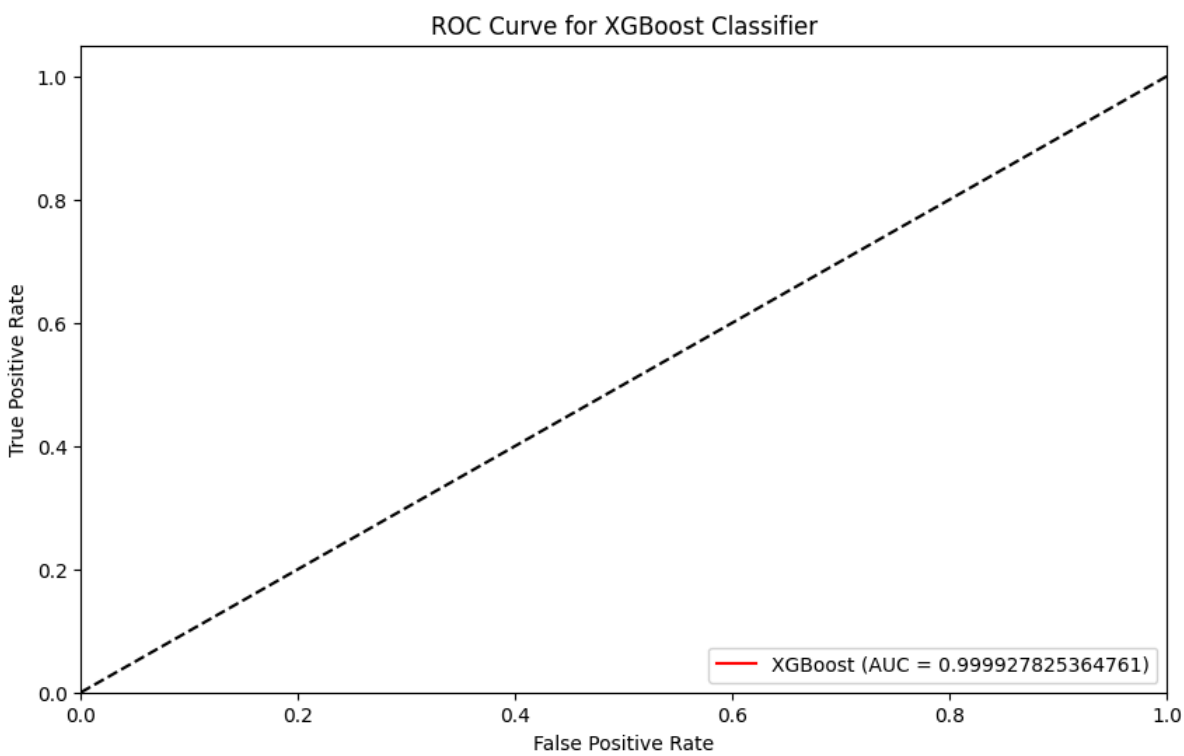
In [85]:
```
from sklearn.metrics import roc_curve, roc_auc_score
y_pred_prob = best_xgb_classifier.predict_proba(X_val)[:,1]
```

Computing ROC curve and AUC Score

In [ ]:
```
fpr, tpr, thresholds = roc_curve(y_val,y_pred_prob, pos_label='A')
auc_score = roc_auc_score(y_val,y_pred_prob)
```

Plotting ROC Curve

In [98]:
```
plt.figure(figsize=(10, 6))
plt.plot(fpr, tpr, label=f'XGBoost (AUC = {auc_score})', color='red')
plt.plot([0, 1], [0, 1], 'k--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve for XGBoost Classifier')
plt.legend(loc='lower right')
plt.show()
```



Reading and Preprocessing Unknown Dataset File

In [71]:
```
data_unknown = pd.read_csv('data_unknown.csv')
data_unknown.head()
```

Out[71]:

	ID	Unnamed: 0	RA_ICRS	DE_ICRS	Source	Plx	PM	pmRA	pmDE	Gı
0	119089	19089	64.085883	46.209497	2.340000e+17	0.3386	1.067	-0.841	-0.656	13.132
1	24912	24912	14.701590	67.754101	5.270000e+17	1.6877	3.527	-1.976	-2.921	12.781
2	6264	6264	55.953330	38.656112	2.240000e+17	0.8378	8.216	6.574	-4.927	12.053
3	81903	81903	91.211868	-14.372262	2.990000e+18	1.1175	12.945	7.306	10.687	11.559
4	110917	10917	75.949263	40.560254	2.010000e+17	0.9456	5.200	3.248	-4.061	15.013

5 rows × 28 columns

In [72]:
```
data_unknown.shape
```

Out[72]:
```
(37135, 28)
```

In [73]:
```
data_unknown.dtypes
```

Out[73]:
```
ID              int64
Unnamed: 0      int64
RA_ICRS       float64
DE_ICRS       float64
Source        float64
Plx           float64
PM            float64
pmRA          float64
pmDE          float64
Gmag          float64
e_Gmag        float64
BPmag         float64
e_BPmag       float64
RPmag         float64
e_RPmag       float64
GRVSmag       float64
e_GRVSmag     float64
BP-RP         float64
BP-G          float64
G-RP          float64
pscol         float64
Teff          float64
Dist          float64
Rad           float64
Lum-Flame     float64
Mass-Flame    float64
Age-Flame     float64
z-Flame       float64
dtype: object
```

Finding Missing Values

In [74]:
```
data_unknown.isnull().sum()
```

Out[74]:
```
ID                0
Unnamed: 0        0
RA_ICRS           0
DE_ICRS           0
Source            0
Plx               0
PM                0
pmRA              0
pmDE              0
Gmag              0
e_Gmag            0
BPmag             0
e_BPmag           0
RPmag             0
e_RPmag           0
GRVSmag       16230
e_GRVSmag     16230
BP-RP             0
BP-G              0
G-RP              0
pscol         35973
Teff              0
Dist              0
Rad               0
Lum-Flame       805
Mass-Flame     3246
Age-Flame      9629
z-Flame         805
dtype: int64
```

Handling Missing Values

Dropping Columns with large number of missing values

In [75]:
```
data_unknown = data_unknown.drop(columns=['GRVSmag','e_GRVSmag','pscol'])
```

Imputing Missing Values with Median Strategy

In [76]:
```
from sklearn.impute import SimpleImputer
impute_columns = ['Lum-Flame','Mass-Flame','Age-Flame','z-Flame']
imputer = SimpleImputer(strategy='median')
data_unknown[impute_columns] = imputer.fit_transform(data_unknown[impute_colum
ns])
```

In [77]:
```
data_unknown.isnull().sum().any()
```

Out[77]:
```
False
```

In [78]:
```
X_unknown = data_unknown.drop(columns=['ID','Unnamed: 0'])
X_unknown = scaler.transform(X_unknown)
```

In [79]:
```
y_pred_unknown = best_model.predict(X_unknown)
y_pred_unknown
```

Out[79]:
```
array([1, 0, 0, ..., 1, 1, 1])
```

Preparing Submission File

Mapping the predicted values to corresponding labels

In [82]:
```
label_mapping = {0: 'A', 1: 'B'}
y_pred_unknown_mapped = [label_mapping[pred] for pred in y_pred_unknown]
```

In [83]:
```
submission_data = pd.DataFrame({'ID': data_unknown['ID'],'SpType-ELS':y_pred_u
nknown_mapped})
submission_data.to_csv('submission.csv', index=False)
```